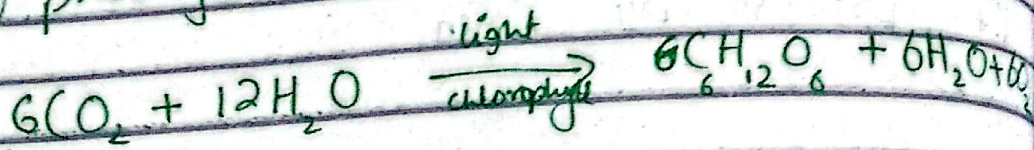


Q: 2(i)

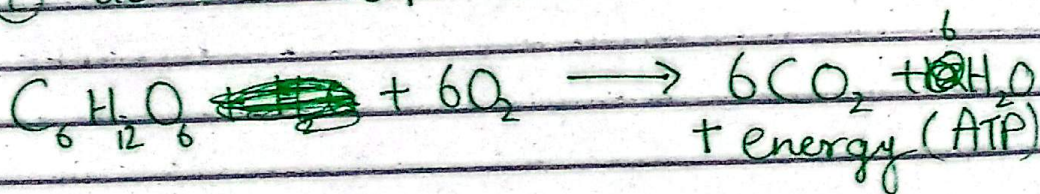
(a) photosynthesis :-



(b) alcoholic fermentation (from pyruvic acid)



(c) aerobic respiration :-



(d)

Lactic acid.

Q: 2(ii)

(a)

Green colour is given by chloroplasts.

(b)

Orange colour is given by chromoplasts.

(c)

Chromoplasts are also found in flower. They give flowers their bright colours to attract pollinators.

(d)

help to attract animals for seed dispersal.

Q:2(iii)

Aerobic

Respiration

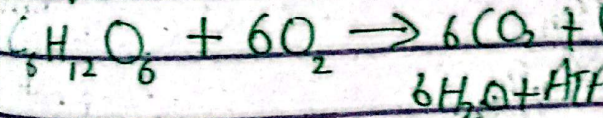
* requires oxygen

* Produces more ATP

* Glucose is completely broken down.

* End products are CO_2 and H_2O

Equation :-



Anaerobic

Respiration.

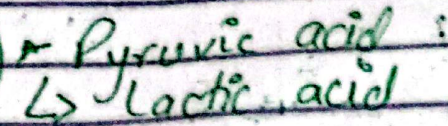
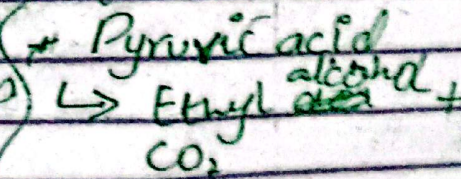
does not requires oxygen.

Produces less ATPs.

Glucose is partially broken down.

End products may be lactic acids or ethanol + CO_2 (yeast)

Equation:-

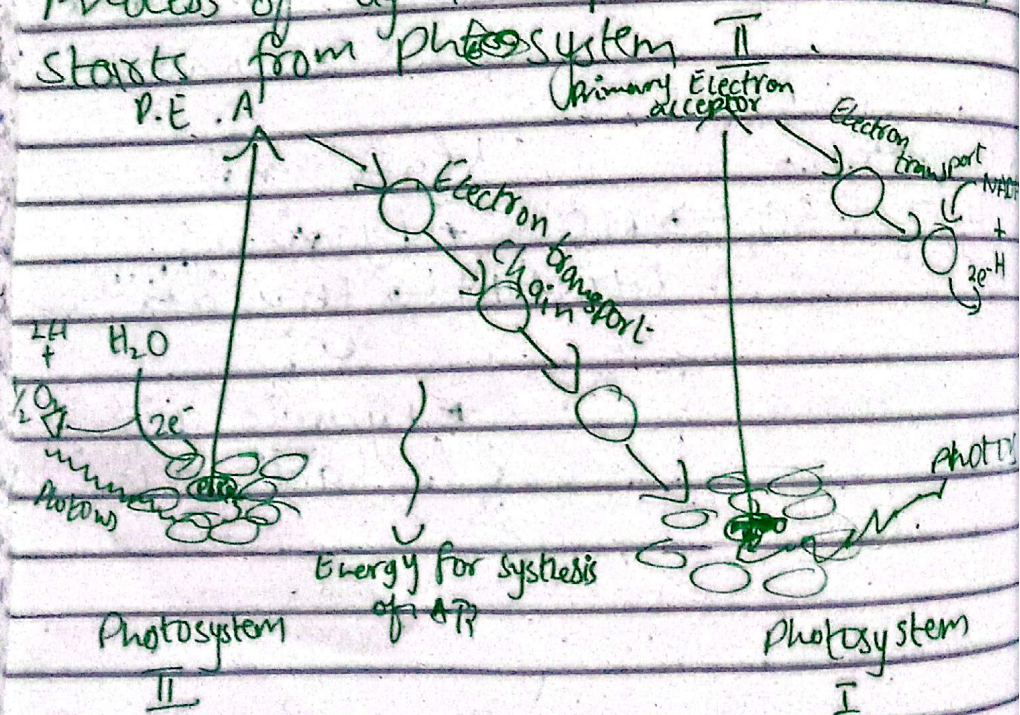


Q.3(i)
Light dependent reaction..

The reactions which depend upon light energy are called light dependent reactions of photosynthesis. These reactions take place in thylakoid membranes where photosynthetic pigments are arranged into clusters called photosystems. There are 2 types of photosystems.

Photosystem I
Photosystem II

Process of light dependent reaction starts from photosystem II.



2) Chlorophyll a of photosystem II absorbs light and a pair of electrons is emitted from it which are accepted by a molecule called primary electron acceptor.

2) At the same time, some water molecules also split into oxygen atom, 2 hydrogen ions and $2e^-$. Splitting of water takes place to compensate the electron loss of light affected chlorophyll of photosystem II and is called photolysis of water.

3) Oxygen is released out of leaf through stomata.

4) e^- pairs from PEA pass through a chain of molecules called electron transport chain (ETC). e^- passing through ETC gradually release energy which is used for ATP synthesis.

3) Light also acts on photosystem I which also gives out an electron pair. These electron pairs and $2H^+$ ions of water

Q.3(i) reduce NADP^+ to NADPH

Q.3(ii)

(a) Prophase II :- Chromosomes condense, the spindle forms and the nuclear envelope breaks down.

2. Metaphase II :- Chromosomes line up at the equator of the cell. Spindle fibres attach to their chromosomes.

3. Anaphase II :- Sister chromatids separate and are pulled to opposite poles by the spindle fibres.

(a) Telophase II :- Chromosomes reach the pole, the nuclear envelope reforms and the chromosomes uncoil.

(b) Similarities

* They both sister chromatids ~~during~~ separate during anaphase

They both have chromosomes
line up at the equator during
metaphase and spindle fibres
are involved

c) Differences

mitosis	meiosis
occurs in somatic cells	occurs in germ line cells
cells divide only once	cells divide twice

MCQs :- e

1. C

reason :- hypothesis is the assumption that comes after questioning & data is a gathered result of research on the biological problem while theory is the well defined statement.

2. D

reason :- theory of observation and hypothesis aren't suitable for this statement's answer.

3. A

reason :- All other options aren't suitable.

4. B

reason :- ~~not~~ other options does not have right ~~answer~~ ^{thing} to answer.

5. B

reason :-

Other options does not make any sense.

6. A

8. ~~AC~~

7. A

9. D

10. B